6.4.3 Quantity Index Number

The quantity index number measures the changes in the level of quantities of items consumed, or produced, or distributed during a year under study with reference to another year known as the base year.

Laspeyre's quantity index:

$$Q_{01}^{L} = \frac{\sum q_1 p_0}{\sum q_0 p_0} \times 100$$

Paasche's quantity index

$$Q_{01}^{P} = \frac{\sum q_1 p_1}{\sum q_0 p_1} \times 100$$

Fisher's quantity index

$$Q_{01}^{F} = \sqrt{Q_{01}^{L} \times Q_{01}^{P}}$$

$$= \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} \times 100$$

These formulae represent the quantity index in which quantities of the different commodities are weighted by their prices.

Example 6.12

Compute the following quantity indices from the data given below:

(i) Laspeyre's quantity index (ii) Paasche's quantity index and (iii) Fisher's quantity index

Commodity	1970		1980	
	Price	Total value	Price	Total value
A	10	80	11	110
B	15	90	9	108
C	8	96	17	340

Solution:

Since we are given the value and the prices, the quantity figures can be obtained by dividing the value by the price for each of the commodities.

Commodity	p_0	q_0	p_1	q_1	p_0q_0	p_1q_0	p_0q_1	p_1q_1
A	10	8	11	10	80	88	100	110
B	15	6	9	12	90	54	180	108
C	8	12	17	20	96	204	160	340
	Total				266	342	440	558

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